

Azul Checkpoint #2



Group 5



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Checkpoint #1 Feedback

Problem 1

The first presentation did not show enough of the filtering process.

We also did not compare flights fairly enough inside similar operating conditions, especially around TOW.

As a result, it was hard to see whether the patterns reflected real operating behavior or sample noise.

Problem 2

Fuel efficiency was still described too broadly and not in an operationally actionable way.

It was difficult to tell exactly what had been done since the first checkpoint and whether the analysis was moving toward implementable route-level decisions.

The recommendations still needed a clearer link between evidence, comparability, and action.

Response: make the filters visible, compare aircraft only in overlap conditions, and move from broad fuel commentary to seat-adjusted route-level evidence.

How we addressed the feedback

1. Show filter impact

Added aircraft-level filtering counts and before-vs-after logic so the audience can see how much data is retained and why.

2. Make fair comparisons

Compared aircraft only inside overlapping TOW conditions, especially E1 vs E2 and A320 vs A321, instead of relying on broad cross-fleet visuals.

3. Define operating windows

Introduced sweetspot ranges based on the best-performing 50% of flights per aircraft to make optimal operating conditions measurable.

4. Move toward action

Added route-overlap and seat-adjusted screening so swap ideas are evaluated on both fuel and approximate capacity fit.

The story is now: clean the data → compare aircraft under similar conditions → define sweetspots → test route-level substitution after capacity screening



Main research question

How can Azul identify implementable aircraft reassignment strategies across route groups with overlapping operational sweet spots in order to improve operational efficiency without assuming full fleet replacement?



Subquestions

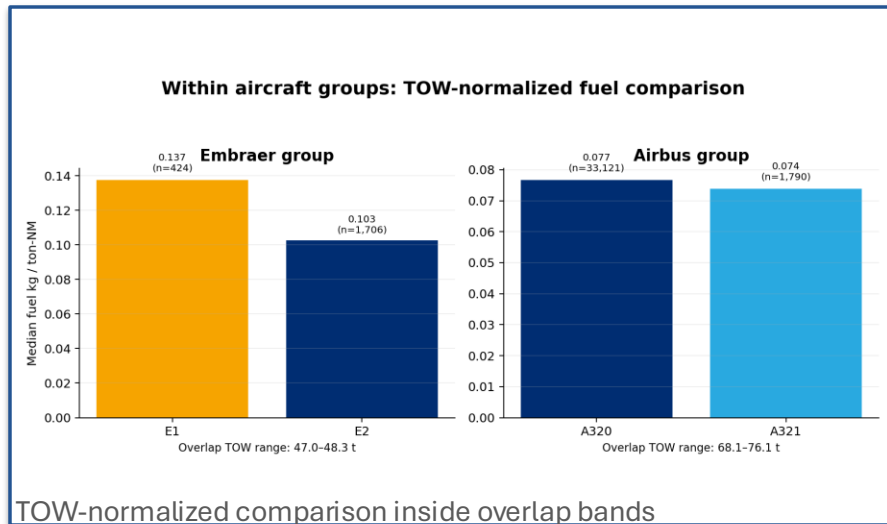
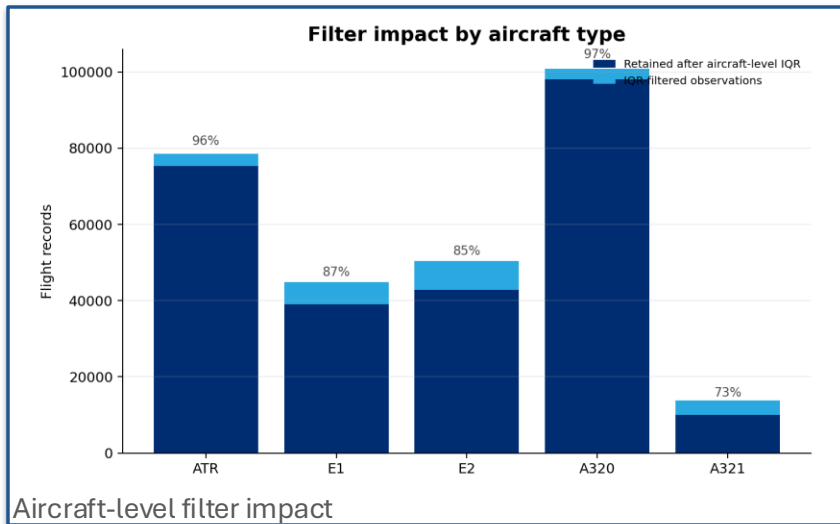
1. Which city pairs or route groups exhibit overlapping operational sweetspots between aircraft types, making reassignment operationally plausible?
2. To what extent do demand fluctuations across routes or seasons justify the reassignment of aircraft types within these overlapping sweetspots?
3. How do airport infrastructure and operational constraints limit the feasibility of using larger or alternative aircraft types on specific routes?
4. How much potential benefit could these reassignment strategies generate for the airline, based on the operational metrics available in the dataset?
5. How sensitive are the identified reassignment opportunities to the analytical assumptions used, particularly seat-based swapability, sweetspot thresholds, and sample size?



Questions based on methodology, validation, or limitations

- Whether payload composition may explain part of the observed differences between aircraft types.
- Whether the selected variables are appropriate for comparing operational fit.
- Whether the analysis contains blind spots or biases.
- Whether seat count is an adequate proxy for swapability.
- Whether the joint sweetspot sample is large enough to support meaningful conclusions.

What we have done since checkpoint #1



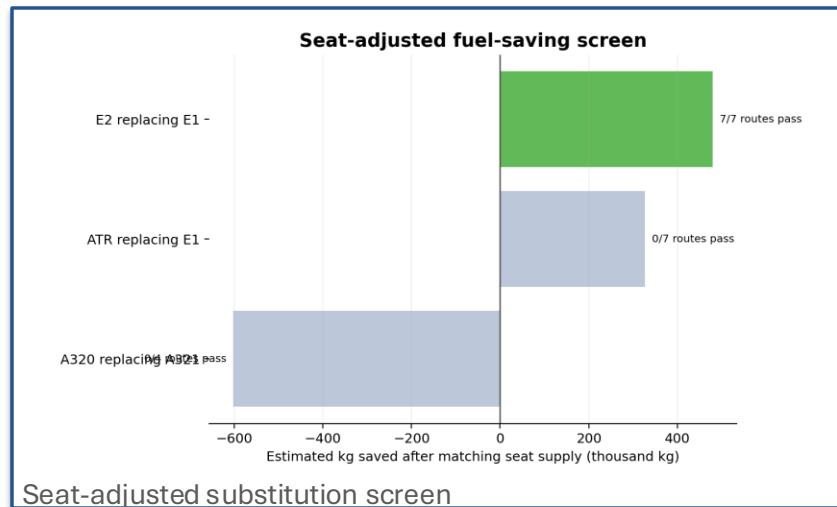
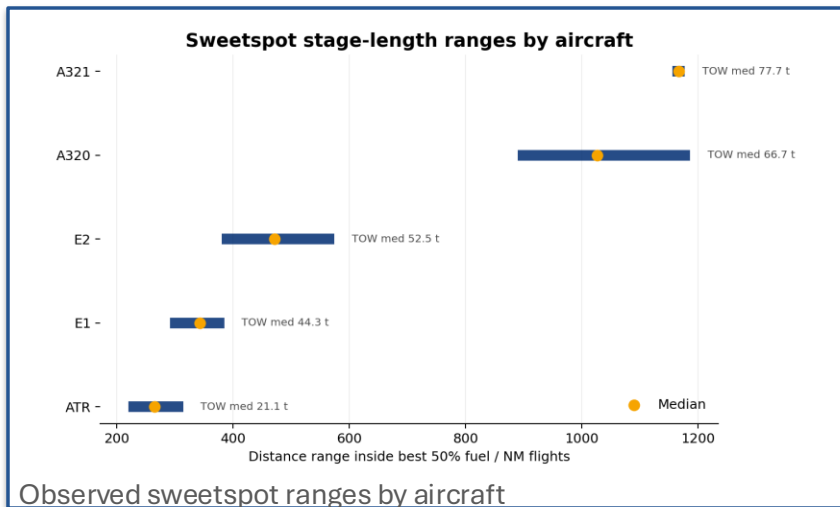
What these additions changed

The IQR filter retains most of the sample while removing noisier tails, especially in E1, E2, and A321.

Inside shared TOW conditions, E2 is clearly more efficient than E1, which gives us a fairer within-family benchmark.

Inside the Airbus overlap, A320 and A321 look much closer, so any swap claim must pass a capacity screen and not rely on fuel-only evidence.

Current findings from the updated notebook



What the current evidence suggests

Observed sweetspots separate aircraft naturally by stage length: ATR 221–321 NM, E1 308–482, E2 442–841, A320 889–1,188, and A321 1,156–1,199.

Fuel-only overlap routes suggest three possible swap directions: A320 replacing A321, ATR replacing E1, and E2 replacing E1.

After matching approximate seat supply, only E2 replacing E1 remains viable across all 7 candidate routes, with about 480 thousand kg of potential savings in the observed overlap sample.

Next analyses and questions for Azul

What we are doing now

- Route-group trade studies inside overlapping sweetspots
- Seasonality by city pair, route type, and stage-length band
- Airport and operating-constraint checks for feasible reassignment
- Theoretical fuel model and scenario-based savings estimation

Questions and data that would help

- Are route additions, removals, or major seasonal shifts already planned?
- Which routes already operate with more than one aircraft type in practice?
- Are there airport, payload, or scheduling constraints that would block reassignment?
- If available: demand/load factor, payload mix, and direct operating cost data

Team ownership by current workstream

Burak

data cleaning, sweetspots, overlap definitions

Robert

comparative visuals, route groups, storyline

Alexandre

constraints, synthesis, Azul-facing recommendations

Eduardo

capacity screen, theoretical model, savings scenarios

RACI matrix project management

Legend
R = Responsible A = Accountable C = Consulted I = Informed

	Burak	Robert	Alexandre	Eduardo
Data foundation and cleaning	R	A	I	C
Checkpoint #1 refinement and visualization updates	C	I	A	R
Deep-dive analyses: sweetspots, route groups, seasonality	A	A	C	R
Optimization modeling: theoretical fuel model, capacity screen, savings	C	R	A	I
Final synthesis: recommendations and presentation story	R	C	C	A

Any questions?



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Ask away!

Thanks for the attention!!